

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278941

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Yu; Ruichi et al.

ASSOCIATION OF CAMERA IMAGES AND RADAR DATA IN AUTONOMOUS VEHICLE APPLICATIONS

Abstract

The described aspects and implementations enable fast and accurate object identification in autonomous vehicle (AV) applications by combining radar data with camera images. In one implementation, disclosed is a method and a system to perform the method that includes obtaining a radar image of a first hypothetical object in an environment of the AV, obtaining a camera image of a second hypothetical object in the environment of the AV, and processing the radar image and the camera image using one or more machine learning models MLMs to obtain a prediction measure representing a likelihood that the first hypothetical object and the second hypothetical object correspond to a same object in the environment of the AV.

Inventors: Yu; Ruichi (Mountain View, CA), Sheng; Shiwei (Cupertino, CA), Li; Kang (Sammamish, WA), Chen; Xu (Livermore, CA)

Applicant: Waymo LLC (Mountain View, CA)

Family ID: 85153357

Appl. No.: 19/211819

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 17444338 20210803 parent-grant-document US 12333817 child US 19211819

Publication Classification

Int. Cl.: G06V20/56 (20220101); G01S7/41 (20060101); G01S13/86 (20060101); G01S13/931 (20200101); G06F18/25 (20230101); G06N20/20 (20190101)

U.S. Cl.:

CPC **G06V20/56** (20220101); **G01S7/417** (20130101); **G01S13/867** (20130101); **G01S13/931** (20130101); **G06F18/251** (20230101); **G06N20/20** (20190101);

Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/444,338, filed Aug. 3, 2021, the contents of which are incorporated by reference in its entirety herein.

TECHNICAL FIELD

[0002] The instant specification generally relates to autonomous vehicles. More specifically, the instant specification relates to improving autonomous driving systems and components using machine learning models to associate camera images with radar data for efficient identification and tracking of objects in autonomous driving environments.

BACKGROUND

[0003] An autonomous (fully and partially self-driving) vehicle (AV) operates by sensing an outside environment with various electromagnetic (e.g., radar and optical) and non-electromagnetic (e.g., audio and humidity) sensors. Some autonomous vehicles chart a driving path through the environment based on the sensed data. The driving path can be determined based on Global Positioning System (GPS) data and road map data. While the GPS and the road map data can provide information about static aspects of the environment (buildings, street layouts, road closures, etc.), dynamic information (such as information about other vehicles, pedestrians, street lights, etc.) is obtained from contemporaneously collected sensing data. Precision and safety of the driving path and of the speed regime selected by the autonomous vehicle depend on timely and accurate identification of various objects present in the driving environment and on the ability of a driving algorithm to process the information about the environment and to provide correct instructions to the vehicle controls and the drivetrain.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is illustrated by way of examples, and not by way of limitation, and can be more fully understood with references to the following detailed description when considered in connection with the figures, in which:

[0005] FIG. 1 is a diagram illustrating components of an example autonomous vehicle capable of combining radar data with camera images for fast and reliable object identification and tracking, in accordance with some implementations of the present disclosure.

[0006] FIG. 2 is a diagram illustrating example architecture of a part of a perception system of an autonomous vehicle that is capable of combining radar data with camera images for fast and reliable object identification and tracking, in accordance with some implementations of the present disclosure.

[0007] FIG. 3A is a schematic illustration of a camera image obtained by a sensing system of an autonomous vehicle and used for identifying correct camera-radar associations, in accordance with some implementations of the present disclosure.

[0008] FIG. 3B is a schematic illustration of three-dimensional radar data obtained by a sensing system of an autonomous vehicle and used for identifying camera-radar associations, in accordance with some implementations of the present disclosure.

[0009] FIG. 4A is a schematic depiction of a projection of three-dimensional radar data onto a two-dimensional surface to obtain projected two-dimensional radar image used for identifying camera-radar associations, in accordance with some implementations of the present disclosure.

[0010] FIG. 4B is a schematic depiction of a combined image obtained by combining camera image with projected two-dimensional radar image, in accordance with some implementations of the present disclosure.

[0011] FIG. 5 is a schematic diagram illustrating an example camera-radar association module that uses one or more machine learning models for fast and reliable object identification and tracking, in accordance with some implementations of the present disclosure.

[0012] FIG. 6A illustrates an example method of forming reliable camera-radar associations in autonomous vehicle applications, in accordance with some implementations of the present disclosure.

[0013] FIG. 6B illustrates example operations of processing radar data and camera image during formation of reliable camera-radar associations, in accordance with some implementations of the present disclosure.

[0014] FIG. 7A illustrates another example method of forming reliable camera-radar associations in autonomous vehicle applications, in accordance with some implementations of the present disclosure.

[0015] FIG. 7B illustrates further example operations of processing radar data and camera image during formation of reliable camera-radar associations, in accordance with some implementations of the present disclosure.

[0016] FIG. 8 depicts a block diagram of an example computer device capable of enabling fast and reliable object identification and tracking in autonomous driving environments, in accordance with some implementations of the present disclosure.

SUMMARY

[0017] In one implementation, disclosed is a method that includes obtaining, by a processing device, a first image, wherein the first image comprises a radar image of a first hypothetical object in an environment of an autonomous vehicle (AV); obtaining, by the processing device, a second image, wherein the second image comprises a camera image of a second hypothetical object in the environment of the AV; and processing the first image and the second image using one or more machine learning models (MLMs) to obtain a prediction measure representing a likelihood that the first hypothetical object and the second hypothetical object correspond to a same object in the environment of the AV.

[0018] In another implementation, disclosed is a method of obtaining, by a processing device, radar data for a region of an environment of an autonomous vehicle (AV); obtaining, by the processing device, a camera image for the region of the environment of the AV; processing the radar data and the camera image using one or more machine learning models (MLMs) to identify an object in the environment of the AV; determining, using the radar data, speed and location of the identified object; and causing a driving path of the AV to be determined in view of the speed and location of the identified object.

[0019] In another implementation, disclosed is a perception system of an autonomous vehicle (AV) to obtain a first image, wherein the first image comprises a radar image of a first hypothetical object in an environment of the AV; obtain a second image, wherein the second image comprises a camera image of a second hypothetical object in the environment of the AV; and process the first image and the second image using one or more machine learning models (MLMs) to obtain a prediction measure representing a likelihood that the first hypothetical object and the second hypothetical object correspond to a same object in the environment of the AV.

DETAILED DESCRIPTION

[0020] An autonomous vehicle can employ a radio detection and ranging (radar) technology and light detection and ranging (lidar) technology to detect distances to various objects in the

environment and the velocities of such objects. A sensor (a radar or lidar) emits one or more radio wave (or laser) signals (e.g., pulses) that travel to an object and then detects arrived signals reflected from the object. By determining a time delay between the signal emission and the arrival of the reflected waves, a sensor can determine the distance to the object. A typical sensor emits signals in multiple directions to obtain a wide view of the outside environment. For example, a sensor (radar or lidar) can cover an entire 360-degree view by using a series of consecutive sensing frames identified with timestamps. As a result, each sector in space is sensed in time increments Δt , which are determined by the angular velocity of the sensor's scanning speed. Sometimes, an entire 360-degree view of the environment can be obtained over a full scan. Alternatively, any smaller sector, e.g., a 1-degree sector, a 5-degree sector, a 10-degree sector, or any other sector can be scanned, as desired.

[0021] Each frame can include numerous return points (or simply “points”) corresponding to reflections from various objects of the environment. Each point can be associated with the distance to the corresponding object or, more specifically, with the distance to an element of the reflective surface of the object (reflecting region) responsible for the respective return point. A set of points within a given frame can be referred to as a “point cloud.” A point cloud can include returns from multiple objects. Typically, it is not known a priori how many objects are within a given frame and to what types (e.g., cars, trucks, buses, motorcycles, pedestrians, etc.) the objects in the frame belong.

[0022] Time-of-flight (ToF) sensors are typically used for ranging. ToFs can also be capable of determining the velocity (speed and direction of motion) of a return point by emitting two or more signals (e.g., as part of different sensing frames) in a quick succession and detecting the position of the reflecting surface as the surface moves with each additional frame. The intervals between successive signals (frames) can be short enough so that the object does not change its position appreciably in relation to other objects of the environment, but still long enough to allow the radar or lidar to detect accurately the changes in the object's position. Coherent sensors, e.g., frequency-modulated continuous wave (FMCW) radars and/or lidars take advantage of a phase information encoded into transmitted signals (and carried by the emitted electromagnetic waves to the target and back) and provide additional functionality. A coherent sensor detects changes in the frequency (and the accompanying phase) of the reflected wave induced by the motion of the reflecting surface, a phenomenon known as the Doppler effect. The frequency/phase of the reflected wave is sensitive to the component of the velocity of the reflecting surface $V_{\text{sub.r}}$ that is parallel to the direction of the wave propagation, herein referred to as the “radial” velocity. A coherent sensor allows to associate a radial velocity with the return points of the point cloud. This additional information (which can be obtained separately or together with the range information) helps the sensing system of the autonomous vehicle to identify the state of the motion of various objects.

[0023] Radar and lidar sensors offer different advantages. A lidar uses waves with a much shorter wavelength than a radar and provides a much higher resolution than radar (the resolution is typically limited by the wavelength). On other hand, radars are less expensive, require less frequent maintenance, have a longer working range of distances, and better tolerate adverse environmental conditions, e.g., foggy, snowy, or misty driving conditions. Because of a significantly lower cost and complexity, more radar devices can be mounted on a vehicle and positioned at locations that (collectively) provide a wider field of view. For example, a view from a single lidar sensor mounted in the middle of the roof of a vehicle may be obscured by an obstacle (e.g., a doubly parked car or truck) whereas at least one of the radar sensors mounted on rear-view mirrors can still have a good view into the obscured portion of the roadway. Radar returns, having a longer wavelength, can often be unable to determine an exact nature of an object that reflects the radar signals, e.g., to differentiate a pedestrian from a road sign, a vehicle from a roadblock, to identify accurate boundaries of the roadway, and so on.

[0024] A camera (e.g., a photographic or video camera) allows a high resolution of objects at both

shorter and longer distances but provides a projection of a three-dimensional space onto a two-dimensional image plane (or some other non-planar surface). A small depiction of an object in a camera image can belong to a small object located close to the camera or to a larger object located at a (proportionally) larger distance. Furthermore, a still camera image can be incapable of informing an autonomous vehicle's perception system as to the speed of the object. A time sequence of camera images can provide the information about the velocity of the object's motion but may require a substantial time for an accurate determination.

[0025] Aspects and implementations of the present disclosure address these and other shortcomings of the existing technologies by enabling methods and systems that reliably and rapidly associate camera data with radar data for faster and more accurate identification and tracking of objects in driving environments. More specifically, camera detection and radar detection have complementary advantages. While camera imaging has a high resolution, radar detection is capable of accurate distance and velocity identification. Challenges of combining camera images with radar data include the need to accurately associate three-dimensional (3D) radar data with two-dimensional (2D) camera outputs. Inaccurate mapping of the two sets of data onto each other may cause misidentifying a stationary close object as moving with a high speed (resulting in an unnecessary emergency braking), or in misidentifying a fast close object as stationary (resulting in a delayed braking), or the like. These challenges are addressed by various implementations of the present disclosure that deploy machine learning models (MLMs) that are trained to accurately and quickly identify various visual depictions in camera images with return points of a radar cloud. In some implementations, an MLM can be trained to process a combined image that includes a camera image with an overlaid radar imaging data and to output a prediction whether an object depicted in the camera image and an object in the radar overlay represent the same object in the driving environment. In some implementations, various additional MLMs can facilitate accurate predictions. For example, another MLM can process the original camera image (without the radar overlay). Yet another MLM can process velocity (Doppler) data obtained by the radar. Outputs (embedding vectors) of the one or more such MLMs can be combined (e.g., concatenated) and processed by a final prediction (classifier) MLM. Advantages of the described implementations include fast and accurate object identification as well as distance/velocity determination. Such an improved perception of autonomous vehicles translates into safer, more comfortable, and more fuel-efficient autonomous driving.

[0026] FIG. 1 is a diagram illustrating components of an example autonomous vehicle (AV) **100** capable of combining radar data with camera images for fast and reliable object identification and tracking, in accordance with some implementations of the present disclosure. Autonomous vehicles can include motor vehicles (cars, trucks, buses, motorcycles, all-terrain vehicles, recreational vehicles, any specialized farming or construction vehicles, and the like), aircraft (planes, helicopters, drones, and the like), naval vehicles (ships, boats, yachts, submarines, and the like), or any other self-propelled vehicles (e.g., robots, factory or warehouse robotic vehicles, sidewalk delivery robotic vehicles, etc.) capable of being operated in a self-driving mode (without a human input or with a reduced human input).

[0027] A driving environment **101** can include any objects (animated or non-animated) located outside the AV, such as roadways, buildings, trees, bushes, sidewalks, bridges, mountains, other vehicles, pedestrians, and so on. The driving environment **101** can be urban, suburban, rural, and so on. In some implementations, the driving environment **101** can be an off-road environment (e.g. farming or other agricultural land). In some implementations, the driving environment can be an indoor environment, e.g., the environment of an industrial plant, a shipping warehouse, a hazardous area of a building, and so on. In some implementations, the driving environment **101** can be substantially flat, with various objects moving parallel to a surface (e.g., parallel to the surface of Earth). In other implementations, the driving environment can be three-dimensional and can include objects that are capable of moving along all three directions (e.g., balloons, leaves, etc.).

Hereinafter, the term “driving environment” should be understood to include all environments in which an autonomous motion of self-propelled vehicles can occur. For example, “driving environment” can include any possible flying environment of an aircraft or a marine environment of a naval vessel. The objects of the driving environment **101** can be located at any distance from the AV, from close distances of several feet (or less) to several miles (or more).

[0028] The example AV **100** can include a sensing system **110**. The sensing system **110** can include various electromagnetic (e.g., optical) and non-electromagnetic (e.g., acoustic) sensing subsystems and/or devices. The sensing system **110** can include a radar **114** (or multiple radars **114**), which can be any system that utilizes radio or microwave frequency signals to sense objects within the driving environment **101** of the AV **100**. The radar(s) **114** can be configured to sense both the spatial locations of the objects (including their spatial dimensions) and velocities of the objects (e.g., using the Doppler shift technology). Hereinafter, “velocity” refers to both how fast the object is moving (the speed of the object) as well as the direction of the object's motion. The sensing system **110** can include a lidar **112**, which can be a laser-based unit capable of determining distances to the objects and velocities of the objects in the driving environment **101**. Each of the lidar **112** and radar **114** can include a coherent sensor, such as a frequency-modulated continuous-wave (FMCW) lidar or radar sensor. For example, radar **114** can use heterodyne detection for velocity determination. In some implementations, the functionality of a ToF and coherent radar is combined into a radar unit capable of simultaneously determining both the distance to and the radial velocity of the reflecting object. Such a unit can be configured to operate in an incoherent sensing mode (ToF mode) and/or a coherent sensing mode (e.g., a mode that uses heterodyne detection) or both modes at the same time. In some implementations, multiple radars **114** and/or lidars **112** can be mounted on AV **100**.

[0029] Radar **114** can include one or more radio/microwave sources producing and emitting signals and one or more detectors of the signals reflected back from the objects. In some implementations, radar **114** (or multiple radars **114**) can perform a 360-degree scanning in a horizontal direction. In some implementations, radar(s) **114** can be capable of spatial scanning along both the horizontal and vertical directions. In some implementations, the field of view can be up to 90 degrees in the vertical direction (e.g., with at least a part of the region above the horizon being scanned by radar signals). In some implementations, the field of view can be a full sphere (consisting of two hemispheres).

[0030] The sensing system **110** can further include one or more cameras **118** to capture images of the driving environment **101**. The images can be two-dimensional projections of the driving environment **101** (or parts of the driving environment **101**) onto a projecting surface (flat or non-flat) of the camera(s). Some of the cameras **118** of the sensing system **110** can be video cameras configured to capture a continuous (or quasi-continuous) stream of images of the driving environment **101**. The sensing system **110** can also include one or more sonars **116**, which can be ultrasonic sonars, in some implementations.

[0031] The sensing data obtained by the sensing system **110** can be processed by a data processing system **120** of AV **100**. For example, the data processing system **120** can include a perception system **130**. The perception system **130** can be configured to detect and track objects in the driving environment **101** and to recognize the detected objects. For example, the perception system **130** can analyze images captured by the cameras **118** and can be capable of detecting traffic light signals, road signs, roadway layouts (e.g., boundaries of traffic lanes, topologies of intersections, designations of parking places, and so on), presence of obstacles, and the like. The perception system **130** can further receive radar sensing data (Doppler data and ToF data) to determine distances to various objects in the environment **101** and velocities (radial and, in some implementations, transverse, as described below) of such objects. In some implementations, the perception system **130** can use radar data in combination with the data captured by the camera(s) **118**, as described in more detail below.

[0032] The perception system **130** can include one or more modules to facilitate association of

radar data with camera data, including one or more camera-radar association models (CRAM) **132** that can be used to process data provided by the sensing system **110**, including images from camera(s) **118** and a radar return points from radar **114**. Input into CRAM **132** can include an image of a portion of the environment depicting one or more objects. Additional input into CRAM **132** can include a radar image (a processed collection of return points) of approximately the same portion of the environment. The radar image can depict an intensity map of the radar signals generated upon reflection from the objects. CRAM **132** can include models that are used to process both types of inputs together to identify correspondence between depictions in the camera image and in the radar image. In some implementations, CRAM **132** can be used to process both images as separate inputs. In some implementations, CRAM **132** can be used to process the images in combination with each other. For example, the radar image can be overlaid over the camera image. In some implementations, CRAM **132** can be used to process a velocity map of the radar returns as an additional input. The velocity map can be a map of the radial (in the direction of detection) velocities of the reflecting objects obtained from a single radar frame. In some implementations, the velocity map can also include lateral velocities of the objects, obtained by tracking the objects between multiple radar frames. Multiple hypothetical pairs of camera/radar images can be input into CRAM **132**, which can identify the best (e.g., most probable) association between camera depictions and radar depictions. The best associations can be identified as specific objects (e.g., vehicles, pedestrians, road signs, buildings or other structures) whose state of motion can subsequently be tracked (e.g., location, velocity, angular velocity, and the like). Various models of CRAM **132** can be trained using multiple camera images and multiple radar images, to identify specific camera and radar features in the respective images associated with the same objects.

[0033] The perception system **130** can further receive information from a positioning subsystem (not shown in FIG. 1), which can include a GPS transceiver (not shown), configured to obtain information about the position of the AV relative to Earth and its surroundings. The positioning subsystem can use the positioning data, e.g., GPS and IMU data) in conjunction with the sensing data to help accurately determine the location of the AV with respect to fixed objects of the driving environment **101** (e.g. roadways, lane boundaries, intersections, sidewalks, crosswalks, road signs, curbs, surrounding buildings, etc.) whose locations can be provided by map information **124**. In some implementations, the data processing system **120** can receive non-electromagnetic data, such as audio data (e.g., ultrasonic sensor data, or data from a mic picking up emergency vehicle sirens), temperature sensor data, humidity sensor data, pressure sensor data, meteorological data (e.g., wind speed and direction, precipitation data), and the like.

[0034] The data processing system **120** can further include an environment monitoring and prediction component **126**, which can monitor how the driving environment **101** evolves with time, e.g., by keeping track of the locations and velocities of the animated objects (e.g., relative to Earth). In some implementations, the environment monitoring and prediction component **126** can keep track of the changing appearance of the environment due to a motion of the AV relative to the environment. In some implementations, the environment monitoring and prediction component **126** can make predictions about how various animated objects of the driving environment **101** will be positioned within a prediction time horizon. The predictions can be based on the current locations and velocities of the animated objects as well as on the tracked dynamics of the animated objects during a certain (e.g., predetermined) period of time. For example, based on stored data for object **1** indicating accelerated motion of object **1** during the previous 3-second period of time, the environment monitoring and prediction component **126** can conclude that object **1** is resuming its motion from a stop sign or a red traffic light signal. Accordingly, the environment monitoring and prediction component **126** can predict, given the layout of the roadway and presence of other vehicles, where object **1** is likely to be within the next 3 or 5 seconds of motion. As another example, based on stored data for object **2** indicating decelerated motion of object **2** during the previous 2-second period of time, the environment monitoring and prediction component **126** can

conclude that object 2 is stopping at a stop sign or at a red traffic light signal. Accordingly, the environment monitoring and prediction component **126** can predict where object 2 is likely to be within the next 1 or 3 seconds. The environment monitoring and prediction component **126** can perform periodic checks of the accuracy of its predictions and modify the predictions based on new data obtained from the sensing system **110**. The environment monitoring and prediction component **126** can operate in conjunction with CRAM **132**. Although not depicted explicitly in FIG. 1, in some implementations, CRAM **132** can be integrated into the environment monitoring and prediction component **126**.

[0035] The data generated by the perception system **130**, the GPS data processing module **122**, and the environment monitoring and prediction component **126** can be used by an autonomous driving system, such as AV control system (AVCS) **140**. The AVCS **140** can include one or more algorithms that control how AV is to behave in various driving situations and environments. For example, the AVCS **140** can include a navigation system for determining a global driving route to a destination point. The AVCS **140** can also include a driving path selection system for selecting a particular path through the immediate driving environment, which can include selecting a traffic lane, negotiating a traffic congestion, choosing a place to make a U-turn, selecting a trajectory for a parking maneuver, and so on. The AVCS **140** can also include an obstacle avoidance system for safe avoidance of various obstructions (rocks, stalled vehicles, a jaywalking pedestrian, and so on) within the driving environment of the AV. The obstacle avoidance system can be configured to evaluate the size of the obstacles and the trajectories of the obstacles (if obstacles are animated) and select an optimal driving strategy (e.g., braking, steering, accelerating, etc.) for avoiding the obstacles.

[0036] Algorithms and modules of AVCS **140** can generate instructions for various systems and components of the vehicle, such as the powertrain, brakes, and steering **150**, vehicle electronics **160**, signaling **170**, and other systems and components not explicitly shown in FIG. 1. The powertrain, brakes, and steering **150** can include an engine (internal combustion engine, electric engine, and so on), transmission, differentials, axles, wheels, steering mechanism, and other systems. The vehicle electronics **160** can include an on-board computer, engine management, ignition, communication systems, carputers, telematics, in-car entertainment systems, and other systems and components. The signaling **170** can include high and low headlights, stopping lights, turning and backing lights, horns and alarms, inside lighting system, dashboard notification system, passenger notification system, radio and wireless network transmission systems, and so on. Some of the instructions output by the AVCS **140** can be delivered directly to the powertrain, brakes, and steering **150** (or signaling **170**) whereas other instructions output by the AVCS **140** are first delivered to the vehicle electronics **160**, which generates commands to the powertrain, brakes, and steering **150** and/or signaling **170**.

[0037] In one example, camera **118**, radar **114**, or lidar can determine that the path ahead is blocked by a doubly parked truck. The AVCS **140** can cause the AV **100** to nudge to the left closer to the median separating the lane occupied by the AV **100** from the oncoming traffic lane. Radar **114** can acquire data that includes a number of returns from an object located within the oncoming traffic lane. Similarly, camera **118** can acquire an image of the region of the oncoming traffic lane that became visible after the nudging maneuver. The camera image and the radar data can be input into CRAM **132** that can identify with an 85% probability that the object in the oncoming traffic lane is a bus. Based on this identification and using the velocity radar data, the perception system **130** can further determine the velocity of the bus to be 38 mph and the bus to be at 110 yards from the doubly parked truck. The data processing system **120** can, therefore, determine that the bus is to pass the doubly parked car in about 6 seconds, the time that is too short to safely drive around the doubly parked truck. Using the determination made by the data processing system **120**, the AVCS **140** can output instructions to powertrain, brakes and steering **150** to stop the AV and wait for the bus to clear the oncoming traffic lane.

[0038] FIG. 2 is a diagram illustrating example architecture **200** of a part of a perception system of an autonomous vehicle that is capable of combining radar data with camera images for fast and reliable object identification and tracking, in accordance with some implementations of the present disclosure. An input into the perception system (e.g., perception system **130** of FIG. 1) can include data obtained by sensing system **110** (e.g., radar **114** and camera **118**), such as distance data, radial velocity data, camera pixel data, etc. For example, a radar data acquisition module **210** can associate each return point with coordinates and radial velocities of the respective reflecting region (e.g., a front bumper of a car, a door of the car, a stop sign, etc.). Various points can correspond to different frames, each frame having a timestamp. A camera image acquisition module **212** can acquire a sequence of images (e.g., similarly identified with timestamps), each image having pixels of various intensities of one color (for black-and-white images) or multiple colors (for color images). An object identification module **220** can process outputs of radar data acquisition module **210** and camera image acquisition module **212**. The data provided to the object identification module **220** can correspond to multiple objects in the environment. Even though directions of camera imaging and radar imaging can be known to a certain degree of precision (e.g., from the timestamps in the respective images and a known camera-radar transceiver synchronization data), exact correspondence between pixels of the camera images and return points of the radar point cloud can be lacking. This can happen because visible light (or infrared) waves used in camera(s) **118** and radio waves (or microwaves) propagate, generally, along somewhat different optical paths. Furthermore, spurious reflections of radio waves (e.g., from buildings, vehicles, bridges, road dividers, etc.) can result in ghost returns from completely different directions. Additionally, radar receivers may have a size that is not significantly larger than the wavelength, so that the accuracy of the angle determination of the received signals may be limited.

[0039] To facilitate accurate camera-radar associations, object identification module **220** can form multiple hypothetical associations of various patches of camera images with various patches of the radar data (e.g., radar reflected intensity data). For example, object identification module **220** can identify N patches segmented from a camera image and depicting a vehicle, a pedestrian, etc. Furthermore, object identification module **220** can identify M possible regions of radar data (e.g., clusters of lidar return points). Object identification module **220** can then identify $N \times M$ possible pairs of camera-radar associations and process each of the identified pairs using CRAM **132**. CRAM **132** can output probabilities that the respective associations are true or false. Object identification module **220** can select the pair of the radar image region j with the camera image region k having the highest probability and can further identify (e.g., based on the depictions in the selected camera and/or radar patches) the type of the object (car, truck, overpass, road sign, bus stop, etc.). CRAM **132** can include one or more MLM, such as decision-tree algorithms, support vector machines, deep neural networks, and the like. Deep neural networks can include convolutional neural networks, recurrent neural networks (RNN) with one or more hidden layers, fully connected neural networks, long short-term memory neural networks, Boltzmann machines, and so on.

[0040] The perception system can include object tracking module **230**, which can receive predicted associations from the object identification module **220**. Object tracking module **230** can also receive (as illustrated by the upper dashed arrow) radar data directly from radar data acquisition module **210**. For example, once a radar blob (cluster of return points) in radar frame A has been identified by CRAM **132**, as corresponding to a motorcycle, object tracking module **230** can monitor subsequent motion of the identified blob using additional radar frames B, C, etc. Tracking can be performed by monitoring coordinates of the blob (e.g., geometric center-of-mass coordinates of the blob, the blob's bounding box, etc.) and velocity (e.g., based on both the radial Doppler-detected velocity and velocity computed from the coordinates).

[0041] Object tracking module **230** can further receive (as illustrated by the lower dashed arrow) additional camera data from camera image acquisition module **212**. Using the coordinates and

velocities determined based on the radar data, object tracking module **230** can verify that the motion of the motorcycle (or any other identified object) inferred from radar data is consistent with the motion of the corresponding object as inferred from the camera images. In some implementations, object tracking module **230** can acquire one of the subsequent radar frames, e.g., frame D with a timestamp $T_{sub.D}$ and segment out a portion of frame D that includes the blob being tracked. Object tracking module **230** can further acquire a camera image with the same timestamp $T_{sub.D}$ (or some other suitable image, e.g., an image with a timestamp that is closest to $T_{sub.D}$) and segment out a patch of the acquired image with the location that approximately corresponds to the tracked radar blob's bounding box. Object tracking module **230** can then provide the portion of the radar frame and the patch of the camera image to CRAM **132** for verification. CRAM **132** can return the probability that the same object is pictured in the two images. If the returned probability is at or above a first threshold probability $p_{sub.1}$ (e.g., 85%, 90%, etc.), object tracking module **230** can determine with high confidence that a correct camera-radar association is still being tracked. Object tracking module **230** can continue tracking the object until the object is beyond a predetermined distance of monitoring (which can depend on the speed and/or type of the identified object and be different for a truck than for a pedestrian) while performing periodic verifications with CRAM **132**, e.g., every n -th radar frame (and/or camera image). If the probability is below the first threshold probability $p_{sub.1}$, but at or above the second threshold probability $p_{sub.2}$ (e.g., 65%, 70%, etc.), object tracking module **230** can determine the status of the tracked object as questionable and place the object for CRAM verification every m -th radar frame (or camera image), where $m < n$. If it is determined by CRAM **132** that the probability is below the second threshold probability $p_{sub.2}$, object tracking module **230** can conclude that the association is no longer valid and stop tracking. In such instances, object identification module **220** can repeat the determination from the beginning, as described above, e.g., by identifying multiple camera patches and multiple radar data regions within a suitable but broadened vicinity of the object previously tracked and then determining the most likely pair.

[0042] Tracking data generated by object tracking module **230** can be provided to AVCS **140**. AVCS **140** evaluates the trajectories of the objects being tracked and determines whether to modify the current driving trajectory of the AV in view of the location and speed of the tracked objects. For example, if a tracked car is within a certain distance from the AV, the AVCS **140** can slow the AV down to a speed that ensures that the car can be safely avoided. Alternatively, AVCS **140** can change lanes, if an adjacent lane is free from obstructions, or perform some other driving maneuver.

[0043] CRAM **132** can be trained using actual camera images and radar data that have been annotated with ground truth, which can include correct camera-radar associations, e.g., based on a human input and/or lidar-based identification. Training can be performed by a training engine **242** hosted by a training server **240**, which can be an outside server that deploys one or more processing devices, e.g., central processing units (CPUs), graphics processing units (GPUs), etc. In some implementations, one or more models of CRAM **132** can be trained by training engine **242** and subsequently downloaded onto the perception system **130** of the AV **100**. CRAM **132**, as illustrated in FIG. 2, can be trained using training data that includes training inputs **244** and corresponding target outputs **246** (correct matches for the respective training inputs). During training of CRAM **132**, training engine **242** can find patterns in the training data that map each training input **244** to the target output **246**.

[0044] Training engine **242** can have access to a data repository **250** storing multiple camera images **252** and instances of radar data **254** for actual driving situations in a variety of environments. During training, training engine **242** can select (e.g., randomly) a number of camera images **252** and radar data **254** as training data. Training data can be annotated with correct camera-radar associations. In some implementations, annotations can be made by a developer before the annotated data is placed into data repository **250**. Annotated training data retrieved by training

server **240** from data repository **250** can include one or more training inputs **244** and one or more target outputs **246**. Training data can also include mapping data **248** that maps training inputs **244** to the target outputs **246**. In some implementations, mapping data **248** can identify an association of a radar blob with an object in a camera image. The mapping data **248** can include an identifier of the training data, location of the object, size of the object, speed and direction of motion of the object, type of the object, and other suitable information.

[0045] During training of CRAM **132**, training engine **242** can change parameters (e.g., weights and biases) of various models of CRAM **132** until the models successfully learn how to predict correct camera-radar associations (target outputs **246**). In some implementations, different models of CRAM **132** (e.g., camera image model, radar data model, etc., as described below in conjunction with FIG. 5) can be trained separately. In some implementations, various models of CRAM **132** can be trained together (e.g., concurrently). Different models can have different architectures (e.g., different numbers of neuron layers and different topologies of neural connections) and, by virtue of separate training, can have different parameters (weights, biases, activation functions, etc.).

[0046] The data repository **250** can be a persistent storage capable of storing images and data as well as data structures configured to identify accurate and fast camera-radar associations, in accordance with implementations of the present disclosure. The data repository **250** can be hosted by one or more storage devices, such as main memory, magnetic or optical storage based disks, tapes or hard drives, NAS, SAN, and so forth. Although depicted as separate from training server **240**, in an implementation, the data repository **250** can be a part of training server **240**. In some implementations, data repository **250** can be a network-attached file server, while in other implementations, data repository **250** can be some other type of persistent storage such as an object-oriented database, a relational database, and so forth, that can be hosted by a server machine or one or more different machines accessible to the training server **240** via a network (not shown in FIG. 2).

[0047] FIG. 3A is a schematic illustration of a camera image **300** obtained by a sensing system of an autonomous vehicle and used for identifying camera-radar associations, in accordance with some implementations of the present disclosure. Camera image **300** can be an individual image obtained by camera **118** of AV **100**, a part of a sequence of images, a single frame of a video obtained by camera **118**, and the like. Camera image **300** can depict cars **302** and **304**, a bus **306**, trees **308** and **310**, a stop sign **312**, and any other objects (not depicted explicitly in FIG. 3A). One or more objects depicted in FIG. 3A can be moving, e.g., car **302**, bus **306**, whereas other objects can be non-moving, e.g., trees **308** and **310**, stop sign **312**, some objects can be stopped temporarily, e.g., car **304**, and so on. Camera image **300** can contain accurate information about angular locations of the objects but can be less informative regarding the distance to the objects and the velocity of the objects.

[0048] FIG. 3B is a schematic illustration of three-dimensional radar data **320** obtained by a sensing system of an autonomous vehicle and used for identifying camera-radar associations, in accordance with some implementations of the present disclosure. Radar data **320** can correspond to camera image **300** of FIG. 3A. Depicted schematically are 3D regions **322** of the environment of an AV that generate radar returns. Although regions **322** are depicted (for conciseness) as rectangular bounding boxes, regions of practically any shape can be identified based on the radar data (e.g., by the radar data acquisition module **210** of FIG. 2). Likewise, although the regions **322** are depicted as continuous, actual physical regions (e.g., bodies and surfaces) that generate the radar returns can be discontinuous, continuous but multiply connected, or can have other possible geometry. 3D regions **322** can include to a single object (e.g., a car), a part of an object, or can span multiple objects (e.g., a car and a nearby stop sign). Each region **322** can be characterized by an intensity of radar returns $I(x, y, z)$, which can depend on a suitably chosen set of 3D coordinates x, y, z , such as Cartesian coordinates, cylindrical coordinates, spherical coordinates, or some other set of coordinates. Each region **322** can be further characterized by a radial velocity $V_{\text{sub.r}}(x, y, z)$. The

intensity $I(x, y, z)$ and the radial velocity $V_{\text{sub.r}}(x, y, z)$ can be defined for a discrete set of coordinates, a quasi-continuous set of coordinates, or for a continuous range of coordinates (e.g., interpolated from a discrete or quasi-continuous set of radar return points).

[0049] FIG. 4A is a schematic depiction of a projection **400** of three-dimensional radar data **320** onto a two-dimensional surface to obtain projected two-dimensional radar image **420** used for identifying camera-radar associations, in accordance with some implementations of the present disclosure. Projection **400** can be on a plane of the camera image **300** (or any other imaging surface of the camera). 3D regions **322** projected on the surface of the camera can have a form of blobs. Some of the blobs can be limited to a single object, e.g., blob **422** can correspond to a car **302** in FIG. 3A and blob **428** can correspond to tree **308**. Some of the blobs can span multiple objects. For example, blob **424** can correspond to closely situated car **304** and stop sign **312** and blob **426** can correspond to bus **306** and tree **310**. Projection **400** can include a projected radar intensity $I(x, y, z)_{\text{fwdarw}}$, $I(x, y)$ and radial velocity $V_{\text{sub.r}}(x, y, z)_{\text{fwdarw}}$, $V_{\text{sub.r}}(x, y)$. In some implementations, one or more models of CRAM **132** can be trained to process camera image **300** and projected 2D radar image **420**. In some implementations, one or more models of CRAM **132** can additionally (or instead) process a combined image that includes both the camera image and the projected 2D radar image **420**.

[0050] FIG. 4B is a schematic depiction of a combined image **450** obtained by combining camera image **300** with projected 2D radar image **420**, in accordance with some implementations of the present disclosure. Combined image can be obtained by overlaying the projected 2D radar image **420** over camera image **300**. In some implementations, overlaying is implemented by adding (weighted) intensity of the radar data $I_{\text{sub.1}}(x, y)$ to the intensity of the camera image $I_{\text{sub.2}}(x, y)$ to obtain the total intensity $w \cdot I_{\text{sub.1}}(x, y) + I_{\text{sub.2}}(x, y)$, where weight w can be selected based on empirical testing. In some implementations, the overlay can be implemented as a two-component intensity vector $\{I_{\text{sub.1}}(x, y), I_{\text{sub.2}}(x, y)\}$. In some implementations, where the camera image is a color image, the vector can be a four-component vector $\{I_{\text{sub.1}}(x, y), I_{\text{sub.R}}(x, y), I_{\text{sub.G}}(x, y), I_{\text{sub.B}}(x, y)\}$, with the intensity of the respective color (Red, Green, and Blue) denoted by a corresponding subscript. In some implementations, the intensity vector can include additional components, e.g., infrared intensity components, provided by the same camera or a separate cameras of the AV.

[0051] FIG. 5 is a schematic diagram illustrating an example camera-radar association module (CRAM) **500** that uses one or more machine learning models for fast and reliable object identification and tracking, in accordance with some implementations of the present disclosure. CRAM **500** can be CRAM **132** of the perception system **130** depicted in FIG. 1. CRAM **500** can include multiple neural networks (NNs), including radar embeddings NN **510**, camera embeddings NN **520**, geometry and motion embeddings NN **530**, and/or other networks not explicitly depicted in FIG. 5. Neurons in the neural networks are associated with learnable weights and biases. The neurons can be arranged in layers. Some of the layers can be hidden layers. Each of the NNs **510**, **520**, and **530** can include multiple neuron layers and can be configured to perform one or more functions that facilitate obtaining correct camera-radar associations. Some of the NNs **510**, **520**, and **530** can be optional.

[0052] The input into CRAM **500** can include a radar data **501** and a camera image **503**. Camera image **503** can be in any suitable digital format (JPEG, TIFF, GIF, BMP, CGM, SVG, and so on). Camera image **503** can be a whole image obtained by a camera of an AV or a patch of the whole image that (prospectively) includes a candidate object being identified, e.g., a vehicle, a road sign, a building, a structure, a pedestrian, or any other object. Similarly, the radar data **501** can include a portion of a radar frame that is hypothesized to correspond to the image patch. In some implementations, the input radar data **501** can be represented as a two-dimensional projection of the three-dimensional radar intensity, as described in more detail in connection with FIG. 4A. In some implementations, the input into the radar embedding NN **510** can be a combined image **502**

that has the radar data **501** overlaid (superimposed) on the camera image **503**, as described in more detail in connection with FIG. **4B**. Input into the geometry and motion embeddings NN **530** can include motion data **504** that can include velocity data and coordinate data. In some implementations, the velocity data can include the radial velocity data obtained from radar Doppler data of a single frame, e.g., using a two-dimensional projection, $V_{\text{sub.r}}(x,y,z) \cdot \text{fwdarw} \cdot V_{\text{sub.r}}(x,y)$. In some implementations, the velocity data can include transverse velocity $V_{\text{sub.}\perp}(x,y)$ that can be obtained from multiple radar frames, e.g., by tracking the motion of a respective radar blob in the lateral direction with the passage of time. In some implementations, the lateral velocity data can be obtained from a series of two or more consecutive camera images. Input into geometry and motion embeddings NN **530** can further include geometry data **505**, such as the coordinate data, e.g., coordinates of the radar blob including but not limited to the coordinates of the bounding box of the radar blob (e.g., corners of the bounding box), a coordinate of a center (e.g., center of mass) of the radar blob, a shape of the surface of the radar blob (e.g., eccentricity, aspect ratio, etc.), and so on. Radar data **501**, camera image **503**, motion data **504**, and geometry data can be obtained and processed in real time.

[0053] Camera image **503** can include a number of pixels. The number of pixels can depend on the resolution of the image, e.g., an image can be represented by 2048×1024 pixels or any other suitable number of pixels. Each pixel can be characterized by one or more intensity values. A black-and-white pixel can be characterized by one intensity value representing the brightness of the pixel, with value 1 corresponding to a white pixel and value 0 corresponding to a black pixel. The intensity value can assume continuous (or discretized) values between 0 and 1 (or between any other chosen limits, e.g., 0 and 255). Similarly, a color pixel can be represented by more than one intensity value, e.g., by three intensity values (e.g., if the RGB color encoding scheme is used) or four intensity values (e.g., if the CMYK color encoding scheme is used). Camera image **503** can be preprocessed prior to being input into camera embeddings NN **520**. For example, camera image **503** can be downsampled (with multiple pixel intensity values combined into a single pixel), upsampled, filtered, denoised, and the like.

[0054] The radar embeddings NN **510** processes combined image **502** and outputs a radar embedding **542**. Similarly, the camera embeddings NN **520** processes camera image **503** and outputs a camera embedding **544**. The geometry and motion embeddings NN **530** processes motion data **504** and outputs a geometry and motion embedding **546**. Each of the NNs **510**, **520**, and **530** can include multiple hidden neuron layers. Processing of input images and data by NNs **510**, **520**, and **530** will next be illustrated for radar embeddings NN **510**, but it should be understood that similar techniques can be used in relation to camera embeddings NN **520** and/or geometry and motion embeddings NN **530**.

[0055] In some implementations, radar embeddings NN **510** can include a number of convolutional layers to extract local and global context of combined image **502**. Convolutional layers can use any suitable parameters, including kernel/mask size, kernel/mask weights, sliding step size, and the like. Convolutional layers can alternate with padding layers and can be followed with one or more pooling layers, e.g., maximum pooling layers, average pooling layers, and the like. Some of the layers of radar embeddings NN **510** can be fully-connected layers. In some implementations, radar embeddings NN **510** can be fully convolutional. The number of nodes in the output layer of radar embeddings NN **510** can be equal to the number of elements of the radar embedding **542**. The radar embedding **542** can be a vector of values that represents the content of combined image **502**. The elements of radar embedding **542** can be an output of nodal operations involving learned weights and biases determined during training.

[0056] Training of radar embeddings NN **510** can be performed using a Siamese network. A Siamese network can be a twin network having the same parameters (weights, biases, and activation functions) as radar embeddings NN **510**. Different images can be input into radar embeddings NN **510** and its Siamese twin network. The embeddings output by radar embeddings

NN **510** and the Siamese twin network can be processed by a classifier network that outputs a binary value, e.g., 1 or 0, indicating whether the input images depict objects of the same type or different types. In some implementations, a type of an object can refer to a high-level classification, such as a car, a semi-truck, a bus, a motorcycle, a pedestrian, a structure, and so on. In some implementations, a type of an object can refer to a medium-level classification, such as a sedan car, an SUV, a sports car, and so on. In some implementations, a type of an object can refer to a low-level classification, that distinguishes different makes of cars, buses, trucks, and the like. A kind of radar embeddings NN **510** to be selected (and trained) for specific uses can be based on the target type of the objects that are intended to be identified (and tracked). For example, radar embeddings NN **510** that is intended to be trained for high-level identification of objects can have fewer number of nodes, hidden layers, etc, than a similar network that is intended to be used for identifying low-level distinctions of objects (and vice versa).

[0057] During training, incorrect classification of different objects as the same or the same objects (or objects of the same type) as different can be used to adjust weights and biases of radar embeddings NN **510** by backpropagating errors (incorrect classifications) through the neuron layers of radar embeddings NN **510**. More specifically, weights and biases of radar embeddings NN **510** are modified until a given pair of training images is successfully recognized. Another pair of images can then be selected and used to further train radar embeddings NN **510**. The training process can continue until a target success rate (e.g., 90%, 95%, etc.) of reliable classification of objects depicted in training images is reached. In some implementations, the target success rate can be determined based on a set of testing images previously not seen by radar embeddings NN **510**. Similar Siamese training can be used for training of other NNs, such as camera embeddings NN **520**, geometry and motion embeddings NN **530**, and/or other NNs that can be deployed as part of CRAM **500**.

[0058] The output of NNs **510-530**, e.g., radar embedding **510**, camera embedding **520**, and geometry and motion embedding **546**, can be joined (e.g., concatenated) into a combined embedding **540**. The combined embedding **540** can be input into a classifier NN **550** trained to output a prediction **560** of the likelihood that the radar data **501** and camera image **503** depict the same object. For example, prediction **560** can be or include a probability, e.g., 20%, 75%, 96%, etc., that the same object is captured by the radar and the camera. Classifier NN **550** can be a fully-connected neural network, in some implementations. In some implementations, classifier NN **550** can include one or more convolutional layers. Classifier NN **550** can be trained separately from NNs **510-530**, e.g., after NNs **510-530** have been successfully trained, as described above. In some implementations, classifier NN **550** can be trained concurrently with NNs **510-530**. In such implementations, Siamese training of NNs **510-530** may not be performed and incorrect predictions **560** can be backpropagated through neuron layers of all NNs **510-530**.

[0059] Prediction **560** can be obtained for multiple pairs of radar data **501** and camera images **503**. With reference to FIG. 2, the obtained predictions **560** can be processed by object identification module **220**, which can select the prediction with the highest probability of association and provide the selected prediction to object tracking module **230** for tracking as described above in connection with FIG. 2. In some implementations, if multiple predictions have probabilities close to the identified maximum probability, e.g., 88%, 84%, 79%, etc. (or within a certain range, e.g., 10%, 15%, etc., from the maximum probability), or if multiple predictions have the probabilities above a certain set threshold (e.g., 85%), object identification module **220** can select all such multiple predictions as temporary hypotheses. Object tracking module **230** can then track the multiple selected predictions as hypotheses for subsequent verification. Each of the tracked hypotheses can be verified by acquiring (using radar data acquisition module **210** and camera image acquisition module **212**) new radar data and new camera image for each of the hypothesized camera-radar associations and processing new data through CRAM **500**. It can be expected that, with time, false associations will become less likely whereas true associations will have probabilities that remain

high (or even increase with time). For example, if two radar blobs in the radar image acquired at time $t_{sub.1}$ are caused by a pedestrian (blob A) walking past a stop sign (blob B) and camera image are similarly cropped into patch a (pedestrian) and patch b (stop sign), CRAM **500** can initially identify four high probability associations (A.Math.a; A.Math.b; B.Math.a; B.Math.b). Object tracking module **230** can then treat all four associations as temporary hypotheses to be tracked. Additional data acquired at a later time $t_{sub.2}$ and input into CRAM **500** can indicate the pedestrian and the stop sign being separated by a larger (than at time $t_{sub.1}$) distance. CRAM **500** can now determine that some associations (e.g., A.Math.a; B.Math.b) still have a high probability (e.g., above a threshold probability or at approximately the same level as at time $t_{sub.1}$) of being correct whereas other associations (e.g., A.Math.b; B.Math.a) have probabilities that are lower than before. Object identification module **220** (or object tracking module **230**) can discard the associations whose probabilities have decreased and continue tracking the remaining associations (e.g., A.Math.a; B.Math.b).

[0060] Multiple variations of the architecture of CRAM can be implemented that differ from CRAM **500** in one or more aspects. In some implementations, motion data **504** is not input into CRAM and no geometry and motion embedding NN **530** is used. In such implementations, combined embedding **540** can include radar embedding **542** and camera embedding **544**. In some implementations, both camera embeddings NN **520** and geometry and motion embeddings NN **530** can be absent and CRAM processed only combined image **502** using radar embeddings NN **510**. For example, the combined image **502** can have the underlying camera image **503** and the radar data **501** overlay represented as separate intensities, e.g., $I_{sub.1}(x, y)$, $I_{sub.2}(x, y)$, etc. The intensities can be processed with filters of one or more initial convolutional layers separately before filters of one or more subsequent convolutional layers begin mixing of separate intensities until the radar embedding **542** is eventually produced, which includes a context of the entire combined image **502**. In some implementations, radar data **501** is input into radar embeddings NN **510** directly, without being combined with the camera image **503**. In such implementations, radar data **501** and camera image **503** are processed separately until classifier NN **550** receives the combined embedding **540**. In some implementations, all NNs being used can be trained together by backpropagating a mismatch between training predictions and target predictions through neuron layers of all NNs of the CRAM.

[0061] FIGS. **6A**, **6B**, **7A**, and **7B** illustrate example methods **600** and **700** and their possible variations of combining radar data with camera images for fast and reliable object identification and tracking. A processing device, having one or more processing units (CPUs) and memory devices communicatively coupled to the CPU(s), can perform methods **600** and **700** and/or each of their individual functions, routines, subroutines, or operations. The processing device executing methods **600** and **700** can perform instructions issued by various components of the perception system **130** of FIG. **1**, e.g., CRAM **132**. Methods **600** and **700** can be directed to systems and components of an autonomous driving vehicle, such as the autonomous vehicle **100** of FIG. **1**. Methods **600** and **700** can be used to improve performance of the autonomous vehicle data processing system **120** and/or the autonomous vehicle control system **140**. In certain implementations, a single processing thread can perform methods **600** and **700**. Alternatively, two or more processing threads can perform methods **600** and **700**, each thread executing one or more individual functions, routines, subroutines, or operations of the methods. In an illustrative example, the processing threads implementing methods **600** and **700** can be synchronized (e.g., using semaphores, critical sections, and/or other thread synchronization mechanisms). Alternatively, the processing threads implementing methods **600** and **700** can be executed asynchronously with respect to each other. Various operations of methods **600** and **700** can be performed in a different order compared with the order shown in FIGS. **6A**, **6B**, **7A**, and **7B**. Some operations of methods **600** and **700** can be performed concurrently with other operations. Some operations can be optional.

[0062] FIG. 6A illustrates an example method **600** of forming reliable camera-radar associations in autonomous vehicle applications, in accordance with some implementations of the present disclosure. At block **610**, method **600** can include obtaining, e.g., by a processing device of the data processing system of an AV, a first image that can include a radar image of a first hypothetical object in an environment of the AV. In some implementations, the radar image can include a three-dimensional representation of the radar data (e.g., intensity $I(x, y, z)$). In some implementations, the radar image can include a two-dimensional projection (e.g., $I(x, y)$) of a three-dimensional radar data obtained for a region of the environment of the AV that includes the first hypothetical object. The first hypothetical object can be any animate or inanimate object in the environment (e.g., a vehicle, a structure, a pedestrian, a bicyclist, a traffic light, a road sign, etc.). In some implementations, the radar image can include additional elements. For example, the radar image can include an overlay of the two-dimensional projection of the three-dimensional radar data onto an image of the environment of the AV taken with a photographic or video camera (referred to herein as a camera image).

[0063] At block **620**, method **600** can continue with the processing device obtaining a second image. The second image can include a camera image of a second hypothetical object in the environment of the AV. The second hypothetical object can be any object in the environment depicted in the second image, which can be the same object as the first hypothetical object or a different object. The second image can be taken of a portion of the environment that includes the first hypothetical object or is proximate to the first hypothetical object but need not be identical to the portion of the environment captured by the radar image.

[0064] At block **630**, method **600** can continue with processing the first image and the second image using one or more machine learning models (MLMs) to obtain a prediction measure. The prediction measure can represent a likelihood that the first hypothetical object and the second hypothetical object correspond to the same object in the environment of the AV. In some implementations, the prediction measure can be a probability that the first and second hypothetical objects are the same. In some implementations, the prediction measure can be a binary value (e.g., 0 or 1, YES or NO, etc.).

[0065] At block **640**, the obtained prediction measure can be used to determine a state of motion of the object. For example, the prediction measure can indicate that the first hypothetical object and the second hypothetical object correspond to the same object. From the camera image, the object can be identified (e.g., using methods of image recognition) as a passenger car having specific attributes (e.g., make, model, color, size, etc.). From the radar image, the state of motion of the object can be determined. The state of motion can include a speed of the object, a location of the object, or both.

[0066] At block **650**, the processing device performing method **600** can cause a driving path of the AV to be determined in view of the state of motion of the object. For example, the perception system of the AV can determine that the object is sufficiently close to the AV (and/or is moving with a substantial speed) to interfere with the driving path of the AV. The perception system can communicate this information to the control system (e.g., the AVCS **140**). The control system can chart a new driving path (or modify the existing driving path) so to avoid the object. The control system can implement the new driving path by outputting instructions to powertrain and steering **150**, vehicle electronics **160**, signaling **170**, etc., to cause the AV to brake, change lanes, stop, back up, accelerate, and so on, and to ensure that the AV follows the determined driving path.

[0067] FIG. 6B illustrates example operations of processing radar data and camera image during formation of reliable camera-radar associations, in accordance with some implementations of the present disclosure. Operations depicted in FIG. 6B can be performed as part of block **630** of method **600** illustrated in FIG. 6A. More specifically, at block **632**, data processing system of AV (e.g., CRAM **132**), can use a first MLM (e.g., radar embeddings NN **510** of FIG. 5) to process the first image to obtain a first embedding vector representing a content of the first image. At block

634, the data processing system can use a second MLM (e.g., camera embeddings NN **520** of FIG. 5) to process the second image to obtain a second embedding vector representing a content of the second image. The first embedding vector and/or a third embedding vector can subsequently be used as an input into a third MLM (e.g., classifier NN **550**).

[0068] At optional (as depicted by the dashed box) block **636**, a fourth MLM (e.g., geometry and motion embeddings NN **530** of FIG. 5) can be used to process the radar data and obtain a third embedding vector representing velocity of the first hypothetical object. For example, the input into the fourth MLM can include velocity (e.g., radial velocity $I(x, y)$) derived from the radar data, e.g., single-frame radar data or multi-frame radar data. More specifically, wherein the radar data can be obtained from two or more radar images of the environment of the AV, each of the images taken at different times and, therefore, capturing temporal changes in the environment. In some implementations, a location (e.g., a bounding box) of the first hypothetical object can be processed by the fourth MLM so that the third embedding vector also represents the location of the object (in addition to the velocity of the object). The third embedding vector can be included into the combined embedding vector (e.g., combined embedding **540**), which can also include (e.g., by concatenation) the first embedding vector and the second embedding vector.

[0069] At block **638**, the processing system of the AV can input the combined embedding vector into the third MLM, to obtain the prediction measure (e.g., prediction **560**). In some implementations, all or at least some of the first MLM, the second MLM, and the fourth MLM can include one or more convolutional neural layers. In some implementations, the third MLM can include one or more fully-connected neuron layers. In some implementations, all or at least some of the first MLM, the second MLM, and the fourth MLM are trained using a Siamese neural network, e.g., a neural network that is identical to the neural network being trained.

[0070] FIG. 7A illustrates another example method **700** of forming reliable camera-radar associations in autonomous vehicle applications, in accordance with some implementations of the present disclosure. At block **710**, method **700** can include obtaining, by a processing device (e.g., one or more CPU of the data processing system **120**), radar data for a region of an environment of an AV. At block **720**, method **700** can continue with the processing device obtaining a camera image for the region of the environment of the AV. The radar data and the camera image do not need to cover the entire region and each can cover a different portion of the region, with the portions having some overlap or (in some instances) having little or no overlap.

[0071] At block **730**, method **700** can continue with the processing device processing the radar data and the camera image using one or more MLMs to identify an object in the environment of the AV. At block **740**, method **700** can continue with determining, using the radar data, speed and location of the identified object, and at block **750** the method can include causing a driving path of the AV to be determined in view of the speed and location of the identified object.

[0072] FIG. 7B illustrates further example operations of processing radar data and camera image during formation of reliable camera-radar associations, in accordance with some implementations of the present disclosure. Operations depicted in FIG. 7B can be performed as part of block **730** of method **700** illustrated in FIG. 7A. More specifically, at block **731**, the data processing system of the AV can obtain a radar image that includes a two-dimensional projection of the radar data on a plane of the camera image. At block **732**, the data processing system can process, using a first MLM, the radar image to obtain a first embedding vector and at block **733** can process, using a second MLM, the camera image to obtain a second embedding vector. At block **734**, the data processing system can process, using a fourth MLM, the radar data to obtain a third embedding vector. The radar data can include the velocity data for one or more objects in the environment of the AV. At block **735**, the data processing system can combine the first embedding vector, the second embedding vector, and the third embedding vector into a combined embedding vector, and can process, at block **736**, the combined embedding vector using a third MLM. The processing by the third MLM can identify an association of a portion of the radar image (e.g., intensity blob

corresponding to a vehicle) and a portion of the camera image (e.g., a picture of the vehicle) with the object (the vehicle) in the environment of the AV.

[0073] FIG. 8 depicts a block diagram of an example computer device **800** capable of enabling fast and reliable object identification and tracking in autonomous driving environments, in accordance with some implementations of the present disclosure, in accordance with some implementations of the present disclosure. Example computer device **800** can be connected to other computer devices in a LAN, an intranet, an extranet, and/or the Internet. Computer device **800** can operate in the capacity of a server in a client-server network environment. Computer device **800** can be a personal computer (PC), a set-top box (STB), a server, a network router, switch or bridge, or any device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that device. Further, while only a single example computer device is illustrated, the term “computer” shall also be taken to include any collection of computers that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methods discussed herein.

[0074] Example computer device **800** can include a processing device **802** (also referred to as a processor or CPU), a main memory **804** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM), etc.), a static memory **806** (e.g., flash memory, static random access memory (SRAM), etc.), and a secondary memory (e.g., a data storage device **818**), which can communicate with each other via a bus **830**.

[0075] Processing device **802** (which can include processing logic **803**) represents one or more general-purpose processing devices such as a microprocessor, central processing unit, or the like. More particularly, processing device **802** can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **802** can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. In accordance with one or more aspects of the present disclosure, processing device **802** can be configured to execute instructions performing methods **600** and **700** of forming reliable camera-radar associations in autonomous vehicle applications.

[0076] Example computer device **800** can further comprise a network interface device **808**, which can be communicatively coupled to a network **820**. Example computer device **800** can further comprise a video display **810** (e.g., a liquid crystal display (LCD), a touch screen, or a cathode ray tube (CRT)), an alphanumeric input device **812** (e.g., a keyboard), a cursor control device **814** (e.g., a mouse), and an acoustic signal generation device **816** (e.g., a speaker).

[0077] Data storage device **818** can include a computer-readable storage medium (or, more specifically, a non-transitory computer-readable storage medium) **828** on which is stored one or more sets of executable instructions **822**. In accordance with one or more aspects of the present disclosure, executable instructions **822** can comprise executable instructions performing methods **600** and **700** of forming reliable camera-radar associations in autonomous vehicle applications.

[0078] Executable instructions **822** can also reside, completely or at least partially, within main memory **804** and/or within processing device **802** during execution thereof by example computer device **800**, main memory **804** and processing device **802** also constituting computer-readable storage media. Executable instructions **822** can further be transmitted or received over a network via network interface device **808**.

[0079] While the computer-readable storage medium **828** is shown in FIG. 8 as a single medium, the term “computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of operating instructions. The term “computer-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of

instructions for execution by the machine that cause the machine to perform any one or more of the methods described herein. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media. [0080] Some portions of the detailed descriptions above are presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0081] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise, as apparent from the following discussion, it is appreciated that throughout the description, discussions utilizing terms such as “identifying,” “determining,” “storing,” “adjusting,” “causing,” “returning,” “comparing,” “creating,” “stopping,” “loading,” “copying,” “throwing,” “replacing,” “performing,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0082] Examples of the present disclosure also relate to an apparatus for performing the methods described herein. This apparatus can be specially constructed for the required purposes, or it can be a general purpose computer system selectively programmed by a computer program stored in the computer system. Such a computer program can be stored in a computer readable storage medium, such as, but not limited to, any type of disk including optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic disk storage media, optical storage media, flash memory devices, other type of machine-accessible storage media, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0083] The methods and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear as set forth in the description below. In addition, the scope of the present disclosure is not limited to any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the present disclosure.

[0084] It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other implementation examples will be apparent to those of skill in the art upon reading and understanding the above description. Although the present disclosure describes specific examples, it will be recognized that the systems and methods of the present disclosure are not limited to the examples described herein, but can be practiced with modifications within the scope of the appended claims. Accordingly, the specification and drawings are to be regarded in an illustrative sense rather than a restrictive sense. The scope of the present disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A method comprising: obtaining, by a processing device, radar data comprising one or more radar depictions of one or more objects in an environment of a vehicle; obtaining, by the processing device, one or more camera images comprising one or more camera depictions of the one or more objects in the environment of the vehicle; processing, using a first machine learning model (MLM), at least the radar data to obtain one or more first embedding vectors associated with the environment of the vehicle; processing, using a second MLM, at least the one or more camera images to obtain a second embedding vector associated with the environment of the vehicle; and determining, using the first embedding vector and the second embedding vector, a prediction measure representing one or more likelihood values, each of the one or more likelihood values representing a likelihood that a radar depiction of the one or more radar depictions and a camera depiction of the one or more camera depictions correspond to a same object in the environment of the vehicle.
2. The method of claim 1, further comprising: controlling, based at least on the prediction measure, a driving path of the vehicle.
3. The method of claim 2, further comprising: determining, in view of the prediction measure, a state of motion of the object, wherein the state of motion comprises at least one of a speed of the object or a location of the object; and wherein controlling the driving path of the vehicle is based on the state of motion of the object.
4. The method of claim 1, wherein the one or more radar depictions and the one or more camera depictions correspond to a same scanning cycle of a sensing system of the vehicle, the sensing system generating the radar data the one or more camera images.
5. The method of claim 1, wherein determining the prediction measure comprises: processing, using a third MLM, a combined embedding vector comprising the one or more first embedding vectors and the one or more second embedding vectors.
6. The method of claim 5, wherein each of the first MLM and the second MLM comprises one or more convolutional neuron layers and wherein the third MLM comprises one or more fully-connected neuron layers.
7. The method of claim 5, wherein determining the prediction measure further comprises: processing, using a fourth MLM, motion data for the one or more objects to obtain one or more third embedding vectors, wherein the motion data comprises one or more of: coordinates of the one or more objects, or velocity of the one or more objects, and wherein the combined embedding vector further comprises the one or more third embedding vectors.
8. The method of claim 1, wherein processing at least the radar data to obtain the one or more first embedding vectors comprises: processing a combined image in which at least one camera image of the one or more camera images is superimposed on the radar data.
9. A non-transitory computer-readable medium storing instructions thereon that, when executed by a processing device, cause the processing device to perform operations comprising: obtaining, by a processing device, radar data comprising one or more radar depictions of one or more objects in an environment of a vehicle; obtaining, by the processing device, one or more camera images comprising one or more camera depictions of the one or more objects in the environment of the vehicle; processing, using a first machine learning model (MLM), at least the radar data to obtain one or more first embedding vectors associated with the environment of the vehicle; processing, using a second MLM, at least the one or more camera images to obtain a second embedding vector associated with the environment of the vehicle; and determining, using the first embedding vector and the second embedding vector, a prediction measure representing one or more likelihood values, each of the one or more likelihood values representing a likelihood that a radar depiction of the one or more radar depictions and a camera depiction of the one or more camera depictions correspond

to a same object in the environment of the vehicle.

10. The non-transitory computer-readable medium of claim 9, wherein the operations further comprise: controlling, based at least on the prediction measure, a driving path of the vehicle.

11. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise: determining in view of the prediction measure, a state of motion of the object, wherein the state of motion comprises at least one of a speed of the object or a location of the object; and wherein controlling the driving path of the vehicle is based on the state of motion of the object.

12. The non-transitory computer-readable medium of claim 9, wherein the one or more radar depictions and the one or more camera depictions correspond to a same scanning cycle of a sensing system of the vehicle, the sensing system generating the radar data the one or more camera images.

13. The non-transitory computer-readable medium of claim 9, wherein determining the prediction measure comprises: processing, using a third MLM, a combined embedding vector comprising the one or more first embedding vectors and the one or more second embedding vectors.

14. The non-transitory computer-readable medium of claim 13, wherein each of the first MLM and the second MLM comprises one or more convolutional neuron layers and wherein the third MLM comprises one or more fully-connected neuron layers.

15. The non-transitory computer-readable medium of claim 13, wherein determining the prediction measure further comprises: processing, using a fourth MLM, motion data for the one or more objects to obtain one or more third embedding vectors, wherein the motion data comprises one or more of: coordinates of the one or more objects, or velocity of the one or more objects, and wherein the combined embedding vector further comprises the one or more third embedding vectors.

16. The non-transitory computer-readable medium of claim 9, wherein processing at least the radar data to obtain the one or more first embedding vectors comprises: processing a combined image in which at least one camera image of the one or more camera images is superimposed on the radar data.

17. A system comprising: a sensing system of a vehicle, to: generate radar data comprising one or more radar depictions of one or more objects in an environment of a vehicle; generate one or more camera images comprising one or more camera depictions of the one or more objects in the environment of the vehicle; and a processing device to: process, using a first machine learning model (MLM), at least the radar data to obtain one or more first embedding vectors associated with the environment of the vehicle; process, using a second MLM, at least the one or more camera images to obtain a second embedding vector associated with the environment of the vehicle; and determine, using the first embedding vector and the second embedding vector, a prediction measure representing one or more likelihood values, each of the one or more likelihood values representing a likelihood that a radar depiction of the one or more radar depictions and a camera depiction of the one or more camera depictions correspond to a same object in the environment of the vehicle.

18. The system of claim 17, wherein the processing device is to further to: determine, in view of the prediction measure, a state of motion of the object, wherein the state of motion comprises at least one of a speed of the object or a location of the object; and controlling a driving path of the vehicle based at least on the state of motion of the object.

19. The system of claim 17, wherein to determine the prediction measure the processing device is to: process, using a third MLM, motion data for the one or more objects to obtain one or more third embedding vectors, wherein the motion data comprises one or more of: coordinates of the one or more objects, or velocity of the one or more objects, and process, using a fourth MLM, a combined embedding vector comprising the one or more first embedding vectors, the one or more second embedding vectors, and the one or more third embedding vectors.

20. The system of claim 17, wherein to process at least the radar data to obtain the one or more first embedding vectors, the processing device is to: process a combined image in which at least one camera image of the one or more camera images is superimposed on the radar data.
